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Original research

A tackler correctly adhering to the tackle instruction in a front-on, one-on-one torso tackle alters the peak inertial head kinematics of the ball carrier but not the tackler

Suzi Edwards [a,*], Andrew J. Gardner [a], Timana Tahu [a], Kenneth L. Quarrie [a,b], Gordon W. Fuller [c], Gary Strangman [d], Grant L. Iverson [e,f,g,h], Ross Tucker [i]

[a] Sydney School of Health Sciences, The University of Sydney, Australia
[b] New Zealand Rugby, New Zealand
[c] Emergency Medicine Research in Sheffield Group, School of Health and Related Research, University of Sheffield, UK
[d] Neural Systems Group, Massachusetts General Hospital, Harvard Medical School, USA
[e] Department of Physical Medicine and Rehabilitation, Harvard Medical School, USA
[f] Spaulding Rehabilitation Hospital and Spaulding Research Institute, USA
[g] MassGeneral Hospital for Children Sport Concussion Program, USA
[h] Home Base, A Red Sox Foundation and Massachusetts General Hospital Program, USA
[i] World Rugby, Pty (Ltd), Ireland

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ABSTRACT

Objectives: To evaluate if the tackler correctly adhering, or not, to four different instructions of legal front-on one-on-one torso tackles altered the tackler and/or ball carrier peak inertial head kinematics.
Design: Controlled laboratory study.
Methods: Fifteen rugby-code players measured with three-dimensional optoelectronic motion capture performed two tackle instructions from the Australian National Rugby League coaching manual on under (*Dominant National Rugby League*) and over (*Smother National Rugby League*) the ball tackles, and two novel variants of these (under, *Dominant, Torso Stick*; over, *Smother, Pop, Lock*). A series of mixed general linear models identified if the tackler adhering ($n = 455$), or not ($n = 139$) to the tackle instructions altered peak inertial head kinematics.
Results: The tackler's peak inertial head kinematics did not significantly change whether or not they adhered to each of the tackle instructions. When the tackler did adhere to the instructions, the ball carrier sustained a lower peak inertial head kinematics ($p < 0.01$) in the *Smother National Rugby League* tackle but higher peak inertial head kinematics in the *Smother, Pop, Lock*.
Conclusions: The ball carriers' inertial head kinematics but not the tacklers were increased when the tackler adhered to this study's variants of the over and under the ball tackle instructions, suggesting that the tacklers were more effective in their tackle performance than the traditional tackle instructions when adhering to the tackle instruction. Greater adherence to the under the ball instructions suggests that the over the ball instruction is a more challenging technique to learn.
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Practical implications

- When performing a 'dominant' tackle by engaging with mid-torso contact (*Dominant, Torso & Stick* [*DTS*]), the tackler should ensure they keep their gaze up and forwards and not flex their trunk too much.
- A tackler executing a 'smother' tackle with the 'pop action' of the *Smother Pop & Lock* (*SPL*) tackle instruction should ensure they lower their body height via a strategy of ankle dorsiflexion, pelvis flexion, and knee flexion.
- Adhering to the *DTS* and the *SPL* tackle instructions increased ball carriers' inertial head kinematics compared to when the tackler does not adhere to the tackle instruction.
- Coaches and tacklers should be aware of the inconsistency of tacklers executing a 'smother' tackle strategy to lower their body position to get under the ball.

* Corresponding author.
E-mail address: suzi.edwards@sydney.edu.au (S. Edwards).
Social media: @Suzi_Edwards (S. Edwards) @TimanaTahu (T. Tahu) @Scienceofsport (R. Tucker).

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1. Introduction

Game play in the rugby codes (i.e., rugby union and rugby league) involves multiple physical collisions between players.[1] The tackle is a physical collision in which the tackler(s) from the defensive team attempt to prevent an attacking player in possession of the ball from the offensive team (i.e., the ball carrier) from advancing down the field and scoring. Some professional rugby league players can perform up to 54 tackles in a game[2] with players on average performing between 11 and 30 tackles per game.[1] In comparison, professional rugby union players execute between 4 and 14 tackles per game.[3]

The risk of concussion during a tackle event is a serious concern, with 95 % of concussions in professional rugby league occurring during the tackle.[4] In professional rugby league, the rate of concussion has been reported at 14.8 concussions per 1000 player match hours[5] (95 % CI 9.6 to 23.0)[6] and 2.9 concussions per 1000 player match hours at a sub-elite level.[7] Similar rates have been reported in professional rugby union with approximately 15.4 (95 % CI 9.5 to 21.3)[8] to 20.4 [9] concussions per 1000 player match hours. The three most common mechanisms that result in head impact events in professional rugby league players are head-to-head, head-to-shoulder, and head-to-hip contact, with the least common contact being head-to-torso.[7,10] Coaching resources have been produced by the rugby code's governing bodies (e.g., the NRL Tackle Ready,[11] RugbySmart,[12] and BokSmart[13]) as an intervention strategy for reducing injury risk in the game (e.g., tackle and/or scrum related). The efficacy of these injury coaching interventions remains unknown[14] and these programmes are often based on expert opinion rather than empirical evidence.[13]

It is critical to provide evidence-based support for any coaching instruction resources. Our previous work has shown that when players adhere to different coaching instructions, the peak resultant inertial head kinematics can be decreased or increased,[15] shoulder flexion and abduction decreased[16] and whole body kinematics of the tackler[17] and ball carrier[18] can be altered. The impact that a tackler's successful execution of certain tackle techniques has on head impulsive forces (i.e., non-direct inertial head kinematics) is not known. Examining the differences in impulsive forces during various tackles requires an accurate measure of the inertial head kinematics of the player. The gold standard for measuring 3D movement is the laboratory-based method of 3D optoelectronic motion capture, as accurate in-game technology does not currently exist.[19] The objective of this study was to evaluate if the tackler correctly adhered, or not, to four different instructions of legal front-on one-on-one torso tackles altered the tackler and/or ball carrier peak inertial head kinematics.

2. Method

2.1. Participants

Healthy and uninjured adult male rugby union or league players (n = 15) were recruited from amateur (club rugby n = 9) or semi-professional (n = 3 regional, n = 2 state, n = 1 international) competition levels. The participant characteristics have been previously reported.[17,18] The study sample size of a minimum of 14 participants was ascertained as adequate based on previous torso tackle research reporting the tackler's trunk flexion at contact.[20] The G*Power software was used to undertake the power calculation using a two-tailed *t*-test with a 0.8 error probability and 95 % statistical power.[21] This pilot study only recruited male participants as it was considered to be important to establish whether tackle instructions executed successfully[15] compared to unsuccessfully executed (i.e., this study's aim) could influence inertial head kinematics in male players, before exploring this concept in female players. This decision to exclude female players was based on the female sex-specific injury risk, particularly breast injury during tackling where it has been shown that 49 % of female football players have sustained a breast injury and 78 % of these breast injuries were caused by contact with another player (49 %) or the ball (29 %).[22] The study design and methodology were approved by the institutional Human Research Ethics Committee (H-2017-0285). All participants provided written, informed consent prior to their participation.

2.2. Experimental procedures

This study adhered to the research framework for undertaking 3D motion tackle analysis outlined by Edwards, Tucker, Quarrie, Tahu, and Gardner.[16] Two participants were involved in every data collection session with a total experimental protocol duration of approximately 3 h. After completing the informed consent and a sporting participation questionnaire, participants had their anthropometric measurements taken. After a full body retroreflective marker set[23] was placed on both participants, a static anatomical standing measurement of both participants was recorded.

Participants, with the guidance of an expert coach, then performed a set of 10 trials for each of the four different front-on one-on-one, torso tackle instructions wearing their own shorts and athletic shoes. Once the 40 tackle trials were completed by the participants (4 sets, 10 trials per set), the participants' roles were reversed (i.e., the ball carrier became the tackler and vice-versa) and another static anatomical position standing measurement was performed, followed by another 40 tackling trials. Rest periods comprised of 30 s between trials, and approximately 5 minute rest between each new technique set of 10 trials.

2.3. Expert coach

The expert coach was a retired dual international rugby league and rugby union player. The expert coach has coaching experience with junior and senior players across both rugby codes. The expert coach developed the non-traditional front-on tackle technique (*SPL*, *DTS*). All sessions were attended by the expert coach. Before any participant commenced the data collection session, the expert coach provided tackling instruction on how to perform the specific technique correctly, as well as guidance and feedback throughout each session. Several familiarisation trials of each tackle instruction were performed with coaching feedback delivered.

The instructions for the two traditional tackle techniques adhered to the NRL coaching manual[24] whereby the tackler's shoulder contacted the lower and upper torso of the ball carrier (i.e., the *Dominant NRL* (*DNRL*) and the *Smother NRL* (*SNRL*) tackle, respectively).[16] Two novel variants of the NRL tackles required the tackler being instructed to modify how they contacted the ball carrier; (i) with a vertical 'pop action' on the mid-upper torso (*SPL*); and (ii) increasing the contact height from lower to mid torso and redirecting the ball carrier into a vertical and backward direction (*DTS*).[16] The tackle instructions of the *DNRL* or *SNRL* were always undertaken first, followed by the two variant tackling techniques (*DTS* or *SPL*). Five dominant and five non-dominant shoulder engagements by the tackler were completed in each set, as evidence is ambiguous on whether the shoulder side engagement alters tackle technique.[25,26]

2.4. Tackling trials

Approach speed was standardised and justification for why the tackle was not taken to the ground was outlined previously by Edwards et al.[17] Participants performed the tackle on a 20 m^2 matted surface (1000 × 1000 × 40 mm jigsaw mat underlay, 2000 × 1000 × 40 mm Tatami Judo mat [Southern Cross Mats, Sydney, NSW & Melbourne, Victoria]).

2.5. Specific tackle instructions

The instructions for the two traditional, front-on one-on-one, torso tackles are outlined in the NRL Coaching Manual.[24] The detailed instructions outlined in these manuals state that the tackler should be as upright as possible when approaching the tackle, then as they are about

to execute the tackle, they are instructed to keep the shoulders higher than the hip level, and to drop their body position whilst they bend their knees. When the tackler contacts the ball carrier's torso the manual outlines that the tackler should use their shoulder, position their head to the ball carrier's side, and keep their neck and spine aligned by ensuring their head is up and straight. When taking the ball carrier to the ground, the tackler is instructed to utilise the ball carrier's momentum and employ strong leg drive.

The tackle height for each of the four tackle instructions included: upper torso (*SNRL*), mid-upper torso (*SPL*), mid torso (*DTS*) and lower torso (*DNRL*), respectively.[15] For the *SPL*, the aim is to redirect the ball carriers' motion into a vertical direction, instead of impeding the horizontal opposing direction of the ball carrier, which is the unique aspect of the *SPL* compared to the *SNRL* instruction. The *SPL* instruction required the tackler to lower their body height just prior to engaging in contact with the ball carrier, then they perform a vertical 'pop action' using their chest/pectoral area to engage in contact with the ball carrier's upper torso. This technique redirects the ball carrier's forward motion into a vertical direction. The two key differences between the *DTS* and *DNRL* instructions are the contact height and the force application angle by the tackler. The *DNRL* instruction requires the tackler to redirect the ball carrier in a backward direction, whereas a backward and vertical direction is instructed in the *DTS*. An increase in torso contact height from lower- to mid-torso is also instructed in the *DTS* to provide the tackler with better positional awareness of the attacking player(s).

2.6. Data collection

The 3D motion of the tackler and ball carrier was captured by 15 Oqus 700+ motion capture cameras (300 Hz, data collection volume 10 m × 10 m × 6 m) using the Qualisys Track Manager software (v.2018.1, Qualisys AB, Göteborg, Sweden) by placing a full body retro-reflective marker set[23] in each player. The Nyquist sampling theorem in this study was adhered to with the appropriate sampling rate ascertained by spectral analysis[27] of 3D head positional data (7 Hz median frequency and 20 Hz mean frequency in which that 99 % of the data occurred below).[16]

Three-dimensional data analysis[16] was performed using the Visual 3D software (Version 6, C-Motion, Germantown, MD, USA). Kinematic variables were calculated using processed data in which a cubic spline interpolated the raw kinematic data that was then filtered by a Butterworth digital filter (zero phase 4th order low pass prior, $f_c = 18$ Hz based on residual analysis[27] of the raw 3D position of head markers with a median frequency of 12 Hz [range 9 to 14 Hz]). Inertial properties and segmental masses were modelled[23] and joint and segment angles were calculated using a local Cartesian coordinate system (x-axis mediolateral; y-axis anterior–posterior; and z-axis superior inferior directions).

The time of contact in the tackle was defined as previously described.[16] At the time of contact joint angles (ankle, knee, hip, lumbopelvic [lumbar segment relative to the pelvis segment], thoracolumbar [lumbar segment relative to the thoracic segment], trunk–pelvis [trunk segment relative to the pelvis segment]), and segment angles (thigh segment, pelvis segment and trunk segment relative to the laboratory coordinate system) of the tackler were reported. Approach speed and speed at contact of each player were defined as the resultant centre of mass velocity at the time of pre-contact (defined as the maximum value prior to contact) and at contact. The foot that made foot–ground contact at Step 1 was defined as the tackler's lead lower limb, and the other foot that made foot–ground contact at Step 2 was defined as the rear lower limb.

2.7. Statistical analysis

Across the 15 participants 594 of the 600 trials collected were included for statistical analysis. A trial was defined as unsuccessful if the tackler did not adhere to the tackling instruction for each specific instruction, which included the tackler's head not being on the side of the ball carriers' body (right shoulder side tackle, left side of ball carriers' body). The determination of adhering or not to the tackle instruction was made by the expert coach and the biomechanics expert together in person by reviewing each trial using the Visual 3D software animation.

Means and standard deviations for successful and unsuccessful trials for every kinematic variable across the four tackling instructions when engaging in the dominant and non-dominant shoulder side were calculated. To ascertain if any significant mean changes ($p < 0.05$) occurred for any outcome variable(s) when the tackling instruction was adhered to, or not a series of general linear mixed models were performing using IBM SPSS (Version 26.0, IBM SPSS Statistics for Windows, Armonk, NY: IBM Corp). There was one repeated measure variable (*instruction*), and four explanatory variables (i.e., subject factors; *adhered to tackle instruction* ($n = 2$), *participant* ($n = 15$), *dominant shoulder side* ($n = 2$) and *count* ($n = 5$)). The 'instruction' explanatory variable signified the four variants of tackle instructions (*DNRL*, *DTS*, *SNRL* and *SPL*). Whether the tackler contacted the ball carrier torso with their dominant or non-dominant shoulder side was defined as the 'dominant shoulder side' explanatory variable. One of the five trials performed per condition (e.g., five trials performed for *SNRL* using the dominant shoulder side) defined the 'count' explanatory variable. To account for the divergent number of trials per person within each condition that the participant did or did not adhere to the tackle instruction, participant and count were also included as explanatory variables.

We confirmed that the normal distribution was an appropriate model for the data using histograms (or normal probability plots) of the model residuals. Regarding assumptions of independence of residuals and constant variance, these were addressed by evaluating models using a range of residual covariance structures (compound, heterogenous compound symmetry and unstructured) and choosing the most appropriate using the Akaike Information Criterion. In this way any correlation between residuals or different variance between conditions was accounted for by the model. Three types of repeated covariance types (compound, heterogenous compound symmetry and unstructured) were performed with each analysis. The model that was selected resulted in the minimisation of the scaling criterion value using the Akaike Information Criterion from each covariance type model.[28] The model with the less complex structure was chosen if the scaling criterion value was <10 between the two models. Main effects of adhered to tackle instruction, dominant shoulder side (i.e., explanatory variables) and repeated measure variable (i.e., four tackle instructions), and their respective two-way and three-way interactions were performed and pairwise contrast types were adjusted for multiple comparisons using least significant difference. The only the statistical results of relevance to the null hypothesis was if adhering to or not the tackle instruction altered the outcome variable. Therefore, only the significant main effect of the explanatory variable of 'adhering to tackle instruction' or a significant interaction between this explanatory variable of 'adhering to tackle instruction' and the repeat measure variable of tackle 'instruction' and/or the explanatory variable of 'dominant shoulder side' was reported.

3. Results

The frequency with which the tackler adhered to the tackle instruction is listed in Table 1. Displayed in Table 2 are the mean ± standard deviations of the peak resultant head accelerations and speed of the tackler and ball carrier, and angles of the tackler across the four tackling instructions when adhering to the tackle instruction or not. Significant main effects and interactions of the mixed general linear model of all variables are shown in Table 3.

Irrespective of whether the tackler executed the tackle adhering to the instruction or not, the tackler and ball carrier instantaneous approach speed, and the instantaneous speed at contact remained constant (Table 2).

Table 1
The number of trials across the four tackling instructions when adhering to the instruction compared to when not.

Instruction	Not adhered		Adhered		Total tackles
	n	%	n	%	n
Dominant Torso & Stick	52	35 %	96	65 %	148
Dominant NRL[a]	1	1 %	148	99 %	149
Smother Pop & Lock	47	32 %	102	68 %	149
Smother NRL[a]	39	26 %	109	74 %	148
Total	139	23 %	455	77 %	594

[a] National Rugby League (NRL) Coaching Manual[23] https://www.playrugbyleague.com/media/3179/nrl-coaching-manual-2019.pdf.

SNRL peak linear and angular head accelerations were significantly lower for the ball carrier when the tackle instruction was adhered to compared when not adhered to (Tables 2, 3). When performed according to the tackle instruction, the *SPL* and *DTS* tackles led to higher peak linear head accelerations for the ball carrier compared to when the instruction was not adhered to.

When adhering to the tackle instruction, the *SPL* resulted in greater more lead and rear dorsiflexion, pelvis flexion, and lumbar flexion when compared to not adhering to the tackle instruction (Tables 2, 3). When adhering to the *DTS* instruction, participants displayed less head flexion, trunk flexion, and trunk–pelvis flexion when executing the tackle on the dominant shoulder side, as opposed to when not adhering to the instruction. The participants correctly adhered to the *SNRL* instruction did so with greater trunk, head, and pelvis flexion, but only when performing the instruction on the dominant shoulder side, compared to the trials when they did not adhere to the tackle instruction.

4. Discussion

This 3D biomechanical study explored the effect that adherence to various front-on, one-on-one, torso tackle instructions has on inertial head kinematics. The results revealed that the successful adherence to the tackle instructions only resulted in the alteration of the inertial head kinematics of the ball carrier compared to when not adhering to the tackle instructions.

Tacklers almost always execute the *DNRL* lower torso tackle instruction correctly, with 99 % of the trials defined as adhering to the tackle instruction. This reflects that this traditionally taught lower torso tackle technique can be easily performed and maintained across numerous tackle efforts and players. The consistent successful execution of this lower torso (i.e., hip area) tackle technique instruction produced the highest peak linear and angular head acceleration for the tackler compared to the mid-torso DSL tackle instruction. It is also known that head-to-hip contact results in the third most common head injury event in professional[10] and youth[29] male rugby league players, with head-to-torso contact the least common. Thus, the magnitude of head impact in the *DNRL* tackle cannot be mitigated by ensuring the tackler adheres to the coaching/technique instruction and also places the tackler at risk of a head-to-hip HIA event.

The lower torso tackle (i.e., *DNRL*) is used by a tackler to execute a 'dominant tackle' by moving the ball carrier backwards and then onto their back by the completion of the tackle.[6] An alternative 'dominant' tackle instruction to the *DNRL* is the *DTS*. Contact to the ball carrier torso in the *DTS* tackle is a slightly higher torso location (mid-torso compared to lower torso) and results in significantly lower peak head linear and angular acceleration than *DNRL*. When the tackler does not adhere to the *DTS* instruction, they were observed to flex their trunk more,

Table 2
Peak resultant head accelerations and speed of the tackler and ball carrier, and angles of the tackler (mean ± SD) across the four tackling instructions when adhering to the instruction compared to when not.

Variable		*Dominant Torso & Stick*		*Dominant NRL*		*Smother Pop & Lock*		*Smother NRL*	
		Not Adhered	Adhered	Not Adhered	Adhered	Not Adhered	Adhered	Not Adhered	Adhered
Peak resultant head acceleration									
Tackler angular ($deg \cdot s^{-2}$)		397 ± 178	321 ± 122	325 ± NA	415 ± 174	344 ± 185	354 ± 172	400 ± 259	352 ± 202
Tackler linear (g)		5.5 ± 2.6	4.6 ± 1.7	6.6 ± NA	6.0 ± 2.2	4.4 ± 1.4	4.1 ± 1.7	5.0 ± 2.2	4.8 ± 2.3
Ball carrier angular ($deg \cdot s^{-2}$)		272 ± 135	282 ± 134	189 ± NA	166 ± 83	285 ± 126	300 ± 143	301 ± 139	214 ± 97[a]
Ball carrier linear (g)		3.8 ± 1.4	4.4 ± 1.7[a]	4.3 ± NA	2.9 ± 1.1	4.4 ± 1.7	5.1 ± 1.8[a]	5.0 ± 1.9	3.8 ± 1.5[a]
Player speed ($m \cdot s^{-1}$)									
Tackler speed at contact		1.8 ± 0.5	1.8 ± 0.5	2.0 ± NA	1.9 ± 0.5	1.9 ± 0.6	2.0 ± 0.5	2.0 ± 0.4	1.8 ± 0.5
Ball carrier speed at contact		2.5 ± 0.6	2.4 ± 0.6	2.7 ± NA	2.3 ± 0.5	2.7 ± 0.4	2.4 ± 0.5	2.5 ± 0.4	2.3 ± 0.6
Tackler peak approach speed		2.4 ± 0.4	2.4 ± 0.4	2.2 ± NA	2.6 ± 0.4	2.4 ± 0.3	2.4 ± 0.4	2.6 ± 0.4	2.5 ± 0.4
Ball carrier peak approach speed		3.2 ± 0.4	3.2 ± 0.4	3.0 ± NA	3.1 ± 0.4	3.2 ± 0.4	3.2 ± 0.4	3.2 ± 0.4	3.1 ± 0.4
Joint angle (°)									
Ankle dorsiflexion	Lead	13.5 ± 11.8	13.6 ± 11.9	17.8 ± NA	12.2 ± 10.9	10.7 ± 15.3	21.7 ± 10.5[a]	10.1 ± 11.4	10.5 ± 9.6
(+)/plantarflexion (−)	Rear	23.0 ± 11.7	19.7 ± 10.9	26.7 ± NA	17.4 ± 13.4	16.9 ± 13.1	22.5 ± 13.0[a]	22.1 ± 14.7	16.9 ± 10.2
Knee flexion (+)/extension (−)	Lead	74.9 ± 14.9	75.2 ± 15.7	82.9 ± NA	76.3 ± 14.4	70.8 ± 13.0	77.6 ± 10.1	73.7 ± 12.9	73.1 ± 12.1
	Rear	81.3 ± 14.4	78.1 ± 14.8	83.6 ± NA	78.5 ± 16.7	74.3 ± 16.4	76.6 ± 16.5	84.0 ± 14.1	81.4 ± 13.2
Hip flexion (+)/extension (−)	Lead	81.8 ± 16.9	77.2 ± 19.8	101.7 ± NA	84.5 ± 18.1	64.6 ± 22.2	65.8 ± 18.7	83.2 ± 22.9	79.7 ± 19.0
	Rear	68.5 ± 17.3	67.7 ± 22.7	89.7 ± NA	68.6 ± 25.9	43.1 ± 17.2	49.6 ± 25.7	73.2 ± 20.2	68.6 ± 25.1
Lumbopelvic flexion (+)/extension (−)		19.4 ± 16.5	23.5 ± 19.9	−3.1 ± NA	23.1 ± 20.3	16.9 ± 15.1	15.6 ± 18.1	14.7 ± 16.4	24.1 ± 20.1
Thoracolumbar flexion (+)/extension (−)		19.7 ± 17.1	16.0 ± 13.8	22.5 ± NA	23.1 ± 16.7	5.4 ± 11.7	6.7 ± 13.8	24.9 ± 19.5	18.9 ± 15.4
Trunk–pelvis extension (+)/flexion (−)		−43.3 ± 12.8	−42.1 ± 16.2[c]	−32.2 ± NA	−49.2 ± 18.0	−26.5 ± 12.1	−25.6 ± 15.4	−41.3 ± 15.0	−46.5 ± 17.4
Segment angle (°)									
Thigh posterior (+)/anterior (−)	Lead	55.0 ± 11.4	55.4 ± 13.1	61.0 ± NA	54.0 ± 13.9	49.4 ± 12.0	48.2 ± 12.9	52.7 ± 19.0	54.9 ± 11.9
inclination	Rear	32.2 ± 20.4	38.4 ± 19.3	46.8 ± NA	30.1 ± 21.7	28.5 ± 21.9	24.2 ± 23.9	38.7 ± 16.7	37.2 ± 20.4
Pelvis extension (+)/flexion (−)		−20.4 ± 17.6	−15.3 ± 23.4	−40.3 ± NA	−21.4 ± 22.9	−7.3 ± 15.3	−15.5 ± 13.9[b,c]	−15.9 ± 25.1	−18.6 ± 18.0[b]
Lumbar extension (+)/flexion (−)		−43.7 ± 12.0	−43.6 ± 10.7	−37.3 ± NA	−49.9 ± 13.3	−28.7 ± 8.8	−32.2 ± 10.0[a,b]	−39.1 ± 9.7	−46.1 ± 11.5
Thoracic extension (+)/flexion (−)		−65.3 ± 20.7	−62.0 ± 13.1	–	−75.2 ± 19.1	−34.9 ± 12.7	−36.5 ± 16.4	−68.8 ± 19.3	−66.3 ± 15.2
Trunk extension (+)/flexion (−)		−68.1 ± 11.0	−62.5 ± 8.9[a]	−72.4 ± NA	−76.8 ± 12.2	−38.7 ± 8.2	−41.0 ± 9.9	−66.1 ± 13.8	−68.5 ± 9.1[a]
Head extension (+)/flexion (−)		−39.4 ± 16.2	−34.7 ± 13.8[a]	−63.3 ± NA	−53.4 ± 18.0	−10.9 ± 14.2	−11.5 ± 12.4	−38.1 ± 15.7	−43.8 ± 17.5[a]

[a] Within instruction factor: when adhering to the tackle instruction it was significantly different than when not adhering to the tackle instruction.
[b] Within dominant shoulder side factor: when adhering to the tackle instruction it was significantly different than when not adhering to the tackle instruction when tackling with their dominant shoulder side only.
[c] Within non-dominant shoulder side factor: when adhering to the tackle instruction it was significantly different than when not adhering to the tackle instruction when tackling with their dominant shoulder side only.

Table 3
General linear mixed model of the main effects and interactions of the tackler across the four tackling instructions for all kinematic variables when adhering to the tackle instruction or not.

Effect	F	p	Post-hoc	p	95 % CI	
					Lower	Upper
Peak resultant head angular acceleration ball carrier						
Instruction * adhering	$F_{3, 557} = 3.7$	0.012	*Smother NRL* lower peak resultant head angular acceleration when adhering to the instruction compared to when not	0.000	28	94
Peak resultant head linear acceleration ball carrier						
Instruction * adhering	$F_{3, 556} = 7.0$	0.000	*Dominant Torso Stick* higher peak resultant head angular acceleration when adhering to the instruction compared to when not	0.031	0.0	0.9
			Smother Pop Lock higher peak resultant head angular acceleration when adhering to the instruction compared to when not	0.017	0.1	1.2
			Smother NRL lower peak resultant head angular acceleration when adhering to the instruction compared to when not	0.001	0.3	1.3
Head extension (+)/flexion (−)						
Instruction * adhering	$F_{3, 574} = 7.6$	0.000	*Dominant Torso Stick* less head flexion when adhering to the instruction compared to when not	0.000	2.98	9.40
			Smother NRL greater head flexion when adhering to the instruction compared to when not	0.007	1.35	8.33
Trunk–pelvis extension (+)/flexion(−)						
Dominant shoulder side * instruction * adhering	$F_{2, 577} = 3.2$	0.043	*Dominant Torso Stick* when tackling with the dominant shoulder side displayed less trunk–pelvis flexion when adhering to the instruction compared to when not	0.033	0.27	6.31
Lead hip flexion (+)/extension(−)						
Dominant shoulder side * adhering	$F_{1, 569} = 4.1$	0.044	–	–	–	–
Lead ankle dorsiflexion (+)/plantarflexion (−)						
Dominant shoulder side * successful	$F_{1, 567} = 5.0$	0.026	–	–	–	–
Instruction * adhering	$F_{3, 567} = 7.5$	0.000	*Smother Pop Lock* greater dorsiflexion when adhering to the instruction compared to when not	0.000	7.55	14.65
Rear ankle dorsiflexion (+)/plantarflexion (−)						
Instruction * adhering	$F_{3, 571} = 4.3$	0.005	*Smother Pop Lock* greater dorsiflexion when adhering to the instruction compared to when not	0.002	1.95	8.38
Rear thigh posterior (+)/anterior (−) inclination						
Dominant shoulder side * adhering	$F_{1, 573} = 3.5$	0.032	–	–	–	–
Pelvis extension (+)/flexion(−)						
Dominant shoulder side * adhering	$F_{1, 575} = 8.0$	0.005	When tackling with the dominant shoulder side, there was greater pelvis flexion when adhering to the instruction compared to when not	0.000	2.9	7.2
Dominant shoulder side * instruction * adhering	$F_{2, 575} = 3.0$	0.050	When tackling with the dominant shoulder side in the *Smother Pop Lock* instruction, there was greater pelvis flexion when adhering to the instruction compared to when not	0.002	2.6	11.6
			When tackling with the dominant shoulder side in the *Smother Pop Lock* instruction, there was greater pelvis flexion when adhering to the instruction compared to when not	0.045	0.1	9.0
			When tackling with the non-dominant compared to dominant shoulder side in the *Smother Pop Lock* instruction, there was less pelvis flexion when adhering to the instruction compared to when not	0.046	0.1	9.1
Lumbar extension (+)/flexion(−)						
Instruction * successful	$F_{3, 575} = 3.6$	0.013	*Smother Pop Lock* greater lumbar flexion when adhering to the instruction compared to when not	0.004	1.1	5.4
Trunk extension (+)/flexion(−)						
Instruction * successful	$F_{3, 575} = 4.1$	0.007	*Dominant Torso Stick* less trunk flexion when adhering to the instruction compared to when not	0.047	0.0	5.6
			Smother NRL greater trunk flexion when adhering to the instruction compared to when not	0.009	0.9	6.3

suggesting the aim of the target is too low (towards the hip area) and they fail to employ an optimal 'head up and forward' gaze, focused on the ball carrier, compared to a successful *DTS* tackle. If the *DTS* instruction aligns better with coaching recommendations[30] and evidence of tackle performance and head[10,30,31] and shoulder[16] injury risk, then the *DTS* tackle instruction should be performed in preference to the *DNRL* when executing a 'dominant tackle' strategy for the tackler. For the tackler, adherence to the *DTS* instruction does not alter the inertial head kinematics, however, it causes the ball carrier to sustain a significantly higher peak resultant linear head acceleration. This high peak inertial head kinematics of the ball carrier likely reflects that the tackler has been more effective in moving the ball carrier backwards in a linear direction. This study did not explore how the ball carrier's behaviour going into the tackle can mitigate their inertial head kinematics. We do know that when the ball carrier does not use an evasion method, their propensity for head injury assessment increases,[7] therefore future research should explore coaching strategies to mitigate their head injury risk when entering a tackle. The highest proportion of unsuccessful trials (46 %) of all of the tackle instructions was attempting to execute the *DTS* tackle. This novel instruction variant was not consistently adhered to, possibly indicating that this technique requires more than subtle changes to execute correctly, which appears to be difficult to achieve when implementing for the first time and with only one coaching session.

The 'smother tackle' is another team's defensive strategy in which the tackler goes over the ball using their chest for contact whilst

wrapping their arms around the ball carrier. The *SPL* instruction significantly reduced the tacklers' peak head linear acceleration compared to the *SNRL*, *DTS*, and *DNRL*. Adhering to either of the smother tackle instructions did not alter the tacklers' inertial head kinematics compared when the instruction was not adhered to. The reduction of the ball carrier's peak inertial head kinetics in the *SNRL* but increasing in the *SPL* suggests that the tacklers were more effective in their tackle performance in the *SPL* but less effective in the *SNRL*.

A consistent inability to execute either of the two smother tackle strategies according to the tackle instructions was observed. Just over half of the over the ball tackle trials were successfully executed by the tackler (53 % for the *SPL*, and 65 % for the *SNRL*) was observed. This inaccuracy in consistently performing an over the ball tackle strategy may be of concern as the two most common mechanisms for a head injury assessment are head-to-head and head-to-shoulder contact,[10] and poorly executed smother tackles place the ball carrier and the tackler in these vulnerable positions. It was anticipated that the *SPL* was likely to have a substantial portion of unsuccessful trials, as participants were learning and attempting to execute this over the ball tackle technique for the first time. However, the finding that participants struggled to consistently perform the traditional upper torso tackle (i.e., the *SNRL*) was unexpected. This may be due to the player's inability to lower their body position to get under the ball and then smother the ball when executing this tackle, or alternatively the players have not been taught this technique according to coaching manuals and instead have developed their own 'hybrid' style of smother tackle. This inconsistency in performing the traditional NRL smother tackle is important for coaches, as whilst the players may have received tackle instructions as a junior player (6–18 years) such as Tackle Ready,[11] the focus on pathway and professional rugby league competitions focuses on defensive systems (e.g., two or three man tacklers) and therefore assumes adult rugby league players' motor skill development of tackling has been correctly learnt and retained during their junior development competition years.

The *SPL* instruction directs the tackler to change how they contacted the ball carrier by engaging in the contact with a vertical 'pop action,' on the mid-upper torso, to go over the ball. When participants successfully adhered to the *SPL* instruction, there is significantly more dorsiflexion of the ankle and more flexion of the pelvis (and trending knee flexion, $p = 0.063$) was seen compared to when they did not successfully perform the tackle instruction. This suggests that when players do not execute the 'pop action' in the *SPL*, it likely occurs as a result of the tackler not lowering their body height via a strategy of ankle dorsiflexion, pelvis flexion, and possible knee flexion. When the tackler follows the *SPL* instruction, it leads to the ball carrier sustaining significantly higher peak resultant linear head acceleration than in an unsuccessful trial, suggesting that the performance of the tackle in moving the ball carrier backwards in a linear direction is more effective.

4.1. Limitations

This preliminary study has several limitations. Risk of bias in the two experts' categorisation of the adherence to tackle instruction is possible. The results of the current study cannot be translated to tackles involving more than one tackler. Variables such as the approach direction of the ball carrier, the player speed, the playing experience or position, the player age, the level of skill, or whether the tackles are completed by taking the ball carrier to ground may limit or prevent the generalisability of the laboratory findings to the real-world match play setting. The robustness of the behavioural modification adherence over longer time duration or the behavioural modification transfer to match play conditions remains unknown. It cannot be determined if previous injury influenced any of the participants' motion as participant injury history was not recorded. The large number of trials ($n = 40$ tackler, $n = 40$ ball carrier) may have caused physical fatigue and/or affected the quality of the performance.

5. Conclusion

Tacklers were most successful at replicating the over the ball *DNRL* tackle instruction, however, this dominant tackle instruction has the highest inertial head kinematics for the tackler but the lowest for the ball carrier. When executing the mid (*DTS*) or mid-upper (*SPL*) torso instruction, the ball carriers' inertial head kinematics were increased, suggesting coaching cue of head and trunk posture and a strategy to lower their body height may improve a tackler's performance and not inadvertently put the ball carrier at greater risk for a head-related injury. The inconsistency in executing an over-the-ball tackle strategy warrants further investigation to identify why tacklers are less able to execute this tackle strategy.

Confirmation of ethical compliance

The study design and methodology were approved by the institutional Human Research Ethics Committee (H-2017-0285). All participants provided written, informed consent prior to their participation.

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Declaration of interest statement

Suzi Edwards declares that they have no conflict of interest. Andrew Gardner, Ph.D. has a clinical practice in neuropsychology involving individuals who have sustained sport-related concussion (including current and former athletes). He has been a contracted concussion consultant to Rugby Australia (2016–). He has received travel funding or been reimbursed by professional sporting bodies, and commercial organisations for discussing or presenting sport-related concussion research at meetings, scientific conferences, workshops, and symposiums. He has received research funding from the National Rugby League (NRL) for the Retired Players Brain Health research programme. Previous grant funding includes the NSW Sporting Injuries Committee, the Brain Foundation (Australia), an Australian–American Fulbright Commission Postdoctoral Award, a Hunter New England Local Health District, Research, Innovation and Partnerships Health Research & Translation Centre and Clinical Research Fellowship Scheme, and the Hunter Medical Research Institute (HMRI), supported by Jennie Thomas, and the HMRI, supported by Anne Greaves. He has served as a scientific advisor for hitIQ Ltd. Ross Tucker is a research consultant to World Rugby (Pty) Ltd, the governing body for Rugby Union globally. Ken Quarrie is employed as the senior sport scientist for New Zealand Rugby. Grant Iverson serves as a scientific advisor for NanoDx™ (formerly BioDirection, Inc.), Sway Operations, LLC, and Highmark, Inc. He has a clinical and consulting practice in forensic neuropsychology, including expert testimony, involving individuals who have sustained mild TBIs (including athletes). He has received research funding from several test publishing companies, including ImPACT Applications, Inc., CNS Vital Signs, and Psychological Assessment Resources (PAR, Inc.). He has received research funding as a principal investigator from the National Football League, and salary support as a collaborator from the Harvard Integrated Program to Protect and Improve the Health of National Football League Players Association Members. Gary Strangman has received pilot research funding from the Harvard Football Player's Health Study, funded through the National Football League Players Association.

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Appendix A. Supplementary data

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